

MYSQL ASSIGNMENT 2

QUERYING DATA

Tasks Clause & Operators:

1. DISTINCT VALUES:

* A query to retrieve distinct salaries from the Employees table.

Table: employees

Columns:
employee_id int PK
Employee_name varchar(100)
Gender char(1)
Age int
Hire_date date
Designation varchar(200)
Salary decimal(10,2)
email varchar(150)
department_id int
location_id int

The screenshot shows a SQL query in a text editor: `SELECT DISTINCT salary FROM employees;`. Below the query, a 'Result Grid' displays the output. The grid has a single column labeled 'salary' and lists 11 distinct salary values: 60000.00, 80000.00, 45000.00, 70000.00, 55000.00, 65000.00, 90000.00, 95000.00, 75000.00, 85000.00, and 100000.00.

salary
60000.00
80000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00
85000.00
100000.00

2. ALIAS (AS):

* Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

The screenshot shows a SQL query in a text editor: `select age as employee_age, salary as employee_salary from employees;`. Below the query, a 'Result Grid' displays the output. The grid has two columns: 'employee_age' and 'employee_salary'. It lists 11 rows of data, each with an age and a salary value.

employee_age	employee_salary
32	60000.00
26	80000.00
33	45000.00
28	70000.00
33	55000.00
32	65000.00
33	90000.00
33	95000.00
33	75000.00
28	85000.00
33	100000.00

3. WHERE CLAUSE & OPERATORS:

* Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

* Find the employee whose designation is missing and fill it with "Data Scientist".

employee_db

- Tables
 - department
 - employees
 - location
- Views
- Stored Procedures
- Functions
- sales_analytics

Administration Schemas

Information

Table: employees

Columns:

Column	Data Type
employee_id	int PK
Employee_name	varchar(100)
Gender	char(1)
Age	int
Hire_date	date
Designation	varchar(200)
Salary	decimal(10,2)
email	varchar(150)
department_id	int
location_id	int

```

90 FROM employees;
91
92 • select
93 age as employee_age,
94 salary as employee_salary
95 from employees;
96
97 • select
98 *
99 from employees
100 where
101 salary > 50000
102 and
103 hire_date < '2016-01-01';
104

```

employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	email	department_id	location_id
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	60000.00	NULL	3	4
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	80000.00	NULL	12	1
5004	Klara Mahotra	F	29	2015-07-05	NULL	70000.00	NULL	8	3
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	55000.00	NULL	11	1
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	65000.00	NULL	8	2

113

```

114 • UPDATE employees
115 SET designation = 'Data Scientist'
116 WHERE designation is NULL;

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

employee_id	Employee_name	Gender	Age	Hire_date	Designation	S
NULL	NULL	NULL	NULL	NULL	NULL	NU

Sorting and Grouping Data:

1. ORDER BY:

Find employees sorted by department ID in ascending order and salary in descending order.

employee_db

- Tables
 - department
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Administration Schemas

Information

Table: employees

Columns:

Column	Data Type
employee_id	int PK
Employee_name	varchar(100)
Gender	char(1)
Age	int
Hire_date	date
Designation	varchar(200)
Salary	decimal(10,2)
email	varchar(150)
department_id	int
location_id	int

```

117
118 • select * from
119 employees
120 order by department_id asc,
121 salary desc;
122
123
124
125
126
127
128
129
130
131

```

employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	email	department_id	location_id
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	90000.00	NULL	1	1
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	90000.00	NULL	2	3
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	70000.00	NULL	2	2
5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	60000.00	NULL	2	1
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	100000.00	NULL	3	4
5026	Vihaan Singh	M	29	2019-03-10	Business Analyst	75000.00	NULL	3	2
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	60000.00	NULL	3	4
5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	100000.00	NULL	4	4
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	60000.00	NULL	4	4
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	55000.00	NULL	4	4
5024	Aryan Shetty	M	27	2018-11-05	Product Manager	95000.00	NULL	5	1
5030	Sara Iyer	F	28	2019-11-20	Project Manager	90000.00	NULL	5	1

2. LIMIT:

Display the first 5 employees hired in the year 2018.

119 • `select * from`
 120 `employees`
 121 `order by department_id asc,`
 122 `salary desc;`
 123
 124 • `select * from`
 125 `employees`
 126 `where year(hire_date) =2018`
 127 `order by hire_date`
 128 `limit 5 ;`

Table: employees
Columns:
 employee_id int PK
 Employee_name varchar(100)
 Gender char(1)
 Age int
 Hire_date date
 Designation varchar(200)
 Salary decimal(10,2)
 email varchar(150)
 department_id int
 location_id int

employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	email	department_id	location_id
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	55000.00	HALS	4	4
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	90000.00	HALS	1	1
5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	85000.00	HALS	7	3
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	60000.00	HALS	4	4
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	70000.00	HALS	2	2

3. AGGREGATE FUNCTIONS:

- Calculate the sum of all salaries in the Finance department.
- Find the minimum age among all employees.

127 `order by hire_date`
 128 `limit 5 ;`
 129
 130 • `select`
 131 `sum(salary) as total_salary`
 132 `from employees`
 133 `where department_id=(`
 134 `select department_id`
 135 `from department`
 136 `where department_name = 'finance'));`

Table: department
Columns:
 department_id int PK
 department_name varchar(100)

total_salary
170000.00

131 `sum(salary) as total_salary`
 132 `from employees`
 133 `where department_id=(`
 134 `select department_id`
 135 `from department`
 136 `where department_name = 'finance'));`
 137
 138 • `select min(age) as min_age from employees;`
 139

Table: employees
Columns:
 employee_id int PK
 Employee_name varchar(100)
 Gender char(1)
 Age int
 Hire_date date
 Designation varchar(200)
 Salary decimal(10,2)
 email varchar(150)
 department_id int
 location_id int

min_age
25

4. GROUP BY:

- List the maximum salary for each location.
- Calculate the average salary for each designation containing the word 'Analyst'.

employee_db

- Tables
 - department
 - employees
 - location
- Views
- Stored Procedures
- Functions
- sales_analytics

Administration Schemas

Information

Table: location

Columns:

- location_id int AI PK
- location_name varchar(100)

```

140 • select location_id,
141       max(salary) as max_salary_location
142   from employees
143  group by location_id;
144
145
146
147
148

```

Result Grid

	location_id	max_salary_location
▶	1	95000.00
	2	95000.00
	3	90000.00
	4	100000.00

employee_db

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- Functions
- sales_analytics

Administration Schemas

Information

Table: location

Columns:

- location_id int AI PK
- location_name varchar(100)

```

144
145 • select * from employees;
146 • select * from location;
147 • select * from department;
148
149 • select DESIGNATION,
150       avg(salary) as avg_salary
151   from employees
152  where designation like '%Analyst%'
153  group by designation;

```

Result Grid

	DESIGNATION	avg_salary
▶	Data Analyst	60000.000000
	Marketing Analyst	60000.000000
	Supply Chain Analyst	60000.000000
	Financial Analyst	85000.000000
	Finance Analyst	85000.000000
	Business Analyst	75000.000000
	Quality Assurance Analyst	80000.000000

5. HAVING:

- Find departments with less than 3 employees.
- Find locations with female employees whose average age is below 30.

employee_db

- Tables
 - department
 - employees
 - location
- Views
- Stored Procedures
- Functions
- sales_analytics

Administration Schemas

Information

Table: employees

Columns:

- employee_id int PK
- Employee_name varchar(100)
- Gender char(1)
- Age int
- Hire_date date
- Designation varchar(200)
- Salary decimal(10,2)
- email varchar(150)
- department_id int
- location_id int

```

154
155
156 • select department_id,
157       count(*) as emp_count
158   from employees
159  group by department_id
160  having emp_count < 3;
161
162

```

Result Grid

	department_id	emp_count
▶	1	1
	5	2
	6	2
	7	2
	9	1
	11	2
	13	1

The screenshot shows a SQL IDE interface. On the left, a tree view displays the database structure for 'employee_db', including tables 'department', 'employees', and 'location'. The 'employees' table is selected. Below the tree, the 'Table: employees' structure is shown with columns: employee_id (int PK), employee_name (varchar(100)), gender (char(1)), age (int), hire_date (date), designation (varchar(200)), salary (decimal(10,2)), email (varchar(150)), department_id (int), and location_id (int). The main editor shows a SQL query:


```

160  having emp_count < 5;
161
162
163  • select location_id,
164     round(avg(age)) as avg_age
165  from
166     employees
167  where gender='f'
168  group by location_id
169  having avg_age < 30;
  
```

 The 'Result Grid' at the bottom shows the following data:

location_id	avg_age
1	27
2	27
4	29

Joins:

1. INNER JOIN:

List employee names, their designations, and department names where employees are assigned to a department.

The screenshot shows a SQL IDE interface. On the left, the 'employee_db' tree view is shown with the 'department' table selected. Below it, the 'Table: department' structure is shown with columns: department_id (int PK) and department_name (varchar(100)). The main editor shows a SQL query:


```

170
171  • select
172     e.employee_name,
173     e.designation,
174     d.department_name
175  from
176     employees e
177  inner join department d
178  on e.department_id = d.department_id;
  
```

 The 'Result Grid' at the bottom displays the following data:

employee_name	designation	department_name
Anvi Chaudhary	Business Development Executive	Business Development
Kiaan Desai	Sales Executive	Business Development
Ananya Paul	Content Writer	Content Creation
Ananya Kapoor	Event Coordinator	Content Creation
Aaradhya Iyer	Customer Support Executive	Customer Support
Arnav Rao	Customer Success Manager	Customer Support
Vihaan Mohan	Supply Chain Analyst	Customer Support
Vihaan Singh	Data Analyst	Data Science
Avni Iyengar	Data Scientist	Data Science
Vivaan Singh	Business Analyst	Data Science
Kiara Malhotra	NULL	Design
Dhruv Shetty	UI Developer	Design
Diya Jha	Graphic Designer	Design
Ishaan Kumar	Financial Analyst	Finance
Advik Kapoor	Finance Analyst	Finance
Zoya Khan	Legal Counsel	Human Resources
Ishika Patel	Talent Acquisition Specialist	Human Resources
Aadhya Iyengar	HR Specialist	Human Resources
Reyansh Singh	Network Engineer	IT
Atharv Yadav	Systems Administrator	IT
Kabir Nair	IT Support Specialist	IT

2. LEFT JOIN:

List all departments along with the total number of employees in each department, including departments with no employees.

Filter objects

- employee_db
 - Tables
 - department
 - employees
 - location
 - Views
 - Stored Procedures
 - Functions
 - sales_analytics

Administration Schemas

Table: **department**

Columns:

- department_id int PK
- department_name varchar(100)

```

179
180 • select
181   d.department_id,
182   d.department_name,
183   count(employee_id) as total_emp
184 from department d
185 left join employees e
186 on d.department_id = e.department_id
187 Group by d.department_id, d.department_name;
188

```

Result Grid

department_id	department_name	total_emp
11	Business Development	2
6	Content Creation	2
10	Customer Support	3
3	Data Science	3
8	Design	3
7	Finance	2
4	Human Resources	3
12	IT	4
2	Marketing	3
13	Operations	1
5	Product Management	2
9	Research and Develo...	1
1	Software Development	1

3. RIGHT JOIN:

Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

Filter objects

- employee_db
 - Tables
 - department
 - employees
 - location
 - Views
 - Stored Procedures
 - Functions
 - sales_analytics

Administration Schemas

Table: **employees**

Columns:

- employee_id int PK
- Employee_name varchar(100)
- Gender char(1)
- Age int
- Hire_date date
- Designation varchar(200)
- Salary decimal(10,2)
- email varchar(150)
- department_id int
- location_id int

```

189
190 • select
191   e.employee_name,
192   l.location_name
193 from employees e
194 right join location l
195 on l.location_id = e.location_id;
196
197
198

```

Result Grid

employee_name	location_name
Aaradhya Iyer	Bangalore
Dhruv Shetty	Bangalore
Myra Verma	Bangalore
Vihaan Mohan	Bangalore
Kabir Nair	Bangalore
Anika Paul	Bangalore
Vivaan Singh	Bangalore
Arjun Kumar	Bangalore
Reyansh Singh	Chennai
Anvi Chaudhary	Chennai
Saanvi Patel	Chennai
Ishaan Kumar	Chennai
Aarav Nair	Chennai
Aryan Shetty	Chennai
Anaya Kapoor	Chennai
Sara Iyer	Chennai
Kiara Malhotra	Hyderabad
Anushka Singh	Hyderabad
Kaia Desai	Hyderabad

4. CROSS JOIN:

Show all possible combinations of departments and locations.

The screenshot shows the SQL Developer interface. On the left, the 'Filter objects' pane shows the 'employee_db' database with tables 'department', 'employees', and 'location'. The 'Schemas' tab is selected. In the center, the SQL editor contains the following query:

```

195
196 • Select
197     d.department_name,
198     l.location_name
199 from department d
200 cross join location l;
201
202
203
204

```

On the right, the 'Result Grid' shows the output of the query:

department_name	location_name
Business Development	Pune
Business Development	Hyderabad
Business Development	Chennai
Business Development	Bangalore
Content Creation	Pune
Content Creation	Hyderabad
Content Creation	Chennai
Content Creation	Bangalore
Customer Support	Pune
Customer Support	Hyderabad
Customer Support	Chennai
Customer Support	Bangalore
Data Science	Pune
Data Science	Hyderabad
Data Science	Chennai
Data Science	Bangalore
Design	Pune
Design	Hyderabad
Design	Chennai
Design	Bangalore

5. SELF JOIN:

Show pairs of employees working in the same department, excluding self-pairs.

The screenshot shows the SQL Developer interface. On the left, the 'Filter objects' pane shows the 'employee_db' database with tables 'department', 'employees', and 'location'. The 'Schemas' tab is selected. In the center, the SQL editor contains the following query:

```

201
202 • select
203     e1.employee_name as emp1,
204     e2.employee_name as emp2,
205     e1.department_id
206 from employees e1
207 join employees e2
208 on e1.department_id = e2.department_id
209 AND e1.employee_id < e2.employee_id;
210

```

On the right, the 'Result Grid' shows the output of the query:

emp1	emp2	department_id
Vihaan Singh	Avni Iyengar	3
Vihaan Singh	Vivaan Singh	3
Reyansh Singh	Atharv Yadav	12
Reyansh Singh	Kabir Nair	12
Reyansh Singh	Arjun Kumar	12
Aaradhya Iyer	Arnav Rao	10
Aaradhya Iyer	Vihaan Mohan	10
Kiara Malhotra	Dhruv Shetty	8
Kiara Malhotra	Diya Jha	8
Anvi Chaudhary	Kiaan Desai	11
Dhruv Shetty	Diya Jha	8
Anushka Singh	Saanvi Patel	2
Anushka Singh	Anika Paul	2
Atharv Yadav	Kabir Nair	12
Atharv Yadav	Arjun Kumar	12
Saanvi Patel	Anika Paul	2
Arnav Rao	Vihaan Mohan	10
Ishaan Kumar	Advik Kapoor	7
Zoya Khan	Ishika Patel	4
Zoya Khan	Aadhya Iyer...	4
Kabir Nair	Arjun Kumar	12

Windows function

- Write a window function query to rank employees by salary using rank().
- Write a window function query to rank employees by salary within each department using DENSE_RANK()
- Write a window function query, Running total salary by department

CHEMAS

Filter objects

employee_db

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Administration Schemas

Information

Table: employees

Columns:

- employee_id int PK
- Employee_name varchar(100)
- Gender char(1)
- Age int
- Hire_date date
- Designation varchar(200)
- Salary decimal(10,2)
- email varchar(150)
- department_id int
- location_id int

```

210
211 • select
212     employee_id,
213     employee_name,
214     salary,
215     rank() over(order by salary desc) as emp_salary
216 from
217 employees;
218

```

Result Grid

employee_id	employee_name	salary	emp_salary
5016	Zoya Khan	100000.00	1
5025	Avni Iyengar	100000.00	1
5012	Myra Verma	95000.00	3
5024	Aryan Shetty	95000.00	3
5007	Anushka Singh	90000.00	5
5020	Aarav Nair	90000.00	5
5030	Sara Iyer	90000.00	5
5015	Ishaan Kumar	85000.00	8
5021	Advik Kapoor	85000.00	8
5002	Reyansh Singh	80000.00	10
5010	Atharv Yadav	80000.00	10
5017	Kabir Nair	80000.00	10
5029	Arjun Kumar	80000.00	10
5013	Arnav Rao	75000.00	14
5018	Ishan Mishra	75000.00	14
5026	Vivran Singh	75000.00	14
5004	Kiara Malhotra	70000.00	17
5008	Diya Jha	70000.00	17
5023	Anika Paul	70000.00	17
5006	Dhruv Shetty	65000.00	20
5001	Vihaan Singh	60000.00	21

CHEMAS

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Table: employees

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- Employee_name varchar(100)
- Gender char(1)
- Age int
- Hire_date date
- Designation varchar(200)
- Salary decimal(10,2)
- email varchar(150)
- department_id int
- location_id int

```

211 • select
212     employee_id,
213     employee_name,
214     department_id,
215     salary,
216     dense_rank() over(partition by department_id order by salary desc) as emp_salary_dep
217 from
218 employees;
219

```

Result Grid

employee_id	employee_name	department_id	salary	emp_salary_dep
5020	Aarav Nair	1	90000.00	1
5007	Anushka Singh	2	90000.00	1
5023	Anika Paul	2	70000.00	2
5011	Saanvi Patel	2	60000.00	3
5025	Avni Iyengar	3	100000.00	1
5026	Vivran Singh	3	75000.00	2
5001	Vihaan Singh	3	60000.00	3
5016	Zoya Khan	4	100000.00	1
5022	Aadhya Iyengar	4	60000.00	2
5019	Ishika Patel	4	55000.00	3
5024	Aryan Shetty	5	95000.00	1
5030	Sara Iyer	5	90000.00	2
5027	Ananya Paul	6	60000.00	1
5028	Anaya Kapoor	6	60000.00	1
5015	Ishaan Kumar	7	85000.00	1
5021	Advik Kapoor	7	85000.00	1
5004	Kiara Malhotra	8	70000.00	1
5008	Diya Jha	8	70000.00	1
5006	Dhruv Shetty	8	65000.00	2
5018	Ishan Mishra	9	75000.00	1
5013	Arnav Rao	10	75000.00	1
5014	Vihaan Mohan	10	60000.00	2

CHEMAS

Filter objects

employee_db

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Table: employees

Columns:

- employee_id int PK
- Employee_name varchar(100)
- Gender char(1)
- Age int
- Hire_date date
- Designation varchar(200)
- Salary decimal(10,2)
- email varchar(150)
- department_id int
- location_id int

```

211 • select
212     employee_id,
213     employee_name,
214     department_id,
215     salary,
216     sum(salary) over(partition by department_id order by employee_id) as emp_salary_total
217 from
218 employees;
219

```

Result Grid

employee_id	employee_name	department_id	salary	emp_salary_total
5020	Aarav Nair	1	90000.00	90000.00
5007	Anushka Singh	2	90000.00	90000.00
5011	Saanvi Patel	2	60000.00	150000.00
5023	Anika Paul	2	70000.00	220000.00
5001	Vihaan Singh	3	60000.00	60000.00
5025	Avni Iyengar	3	100000.00	160000.00
5026	Vivran Singh	3	75000.00	235000.00
5016	Zoya Khan	4	100000.00	100000.00
5019	Ishika Patel	4	55000.00	155000.00
5022	Aadhya Iyengar	4	60000.00	215000.00
5024	Aryan Shetty	5	95000.00	95000.00
5030	Sara Iyer	5	90000.00	185000.00
5027	Ananya Paul	6	60000.00	60000.00
5028	Anaya Kapoor	6	60000.00	120000.00
5015	Ishaan Kumar	7	85000.00	85000.00
5021	Advik Kapoor	7	85000.00	170000.00
5004	Kiara Malhotra	8	70000.00	70000.00
5006	Dhruv Shetty	8	65000.00	135000.00
5008	Diya Jha	8	70000.00	205000.00
5018	Ishan Mishra	9	75000.00	75000.00
5003	Aaradhya Iyer	10	45000.00	45000.00
5013	Arnav Rao	10	75000.00	120000.00
5014	Vihaan Mohan	10	60000.00	180000.00